

Engineering Agent Loops and Graphs with Local Models

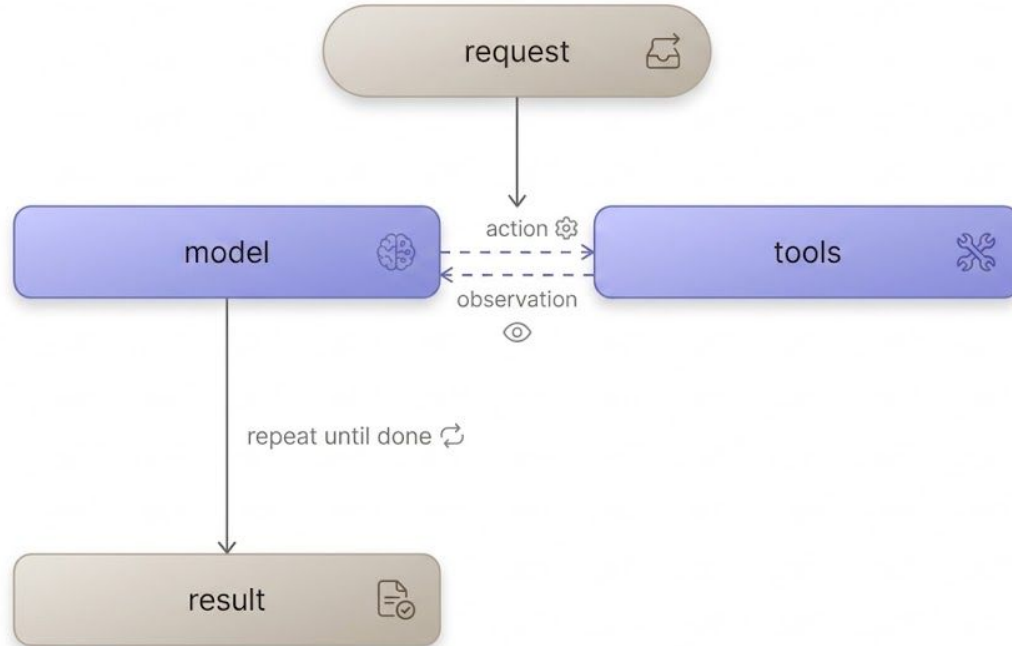
Dustin Dannenhauer
Delegance AI

Outline

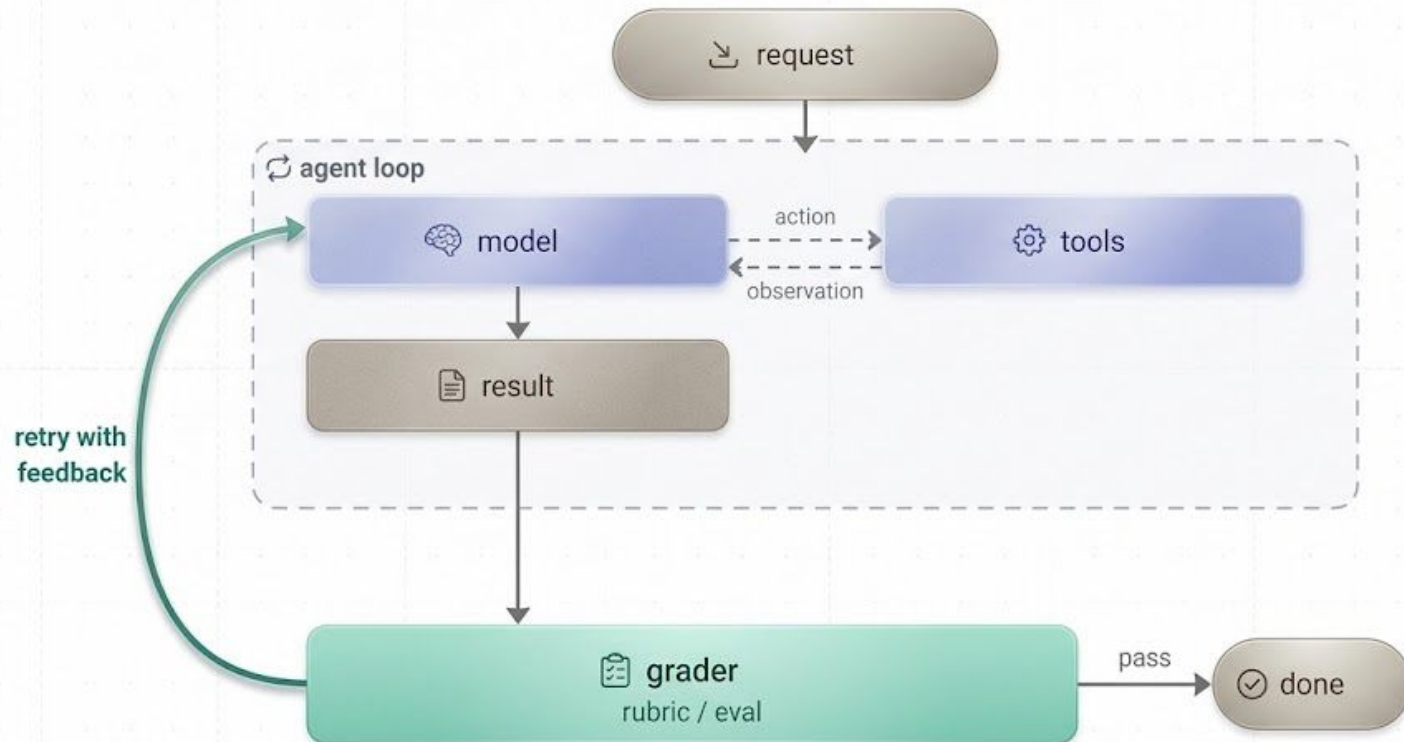
1. **Loops, Graphs, and Workflows**
2. General Approaches to Agentic Coding Work
3. Software as a Discovery Process
4. Early look at Playbooks

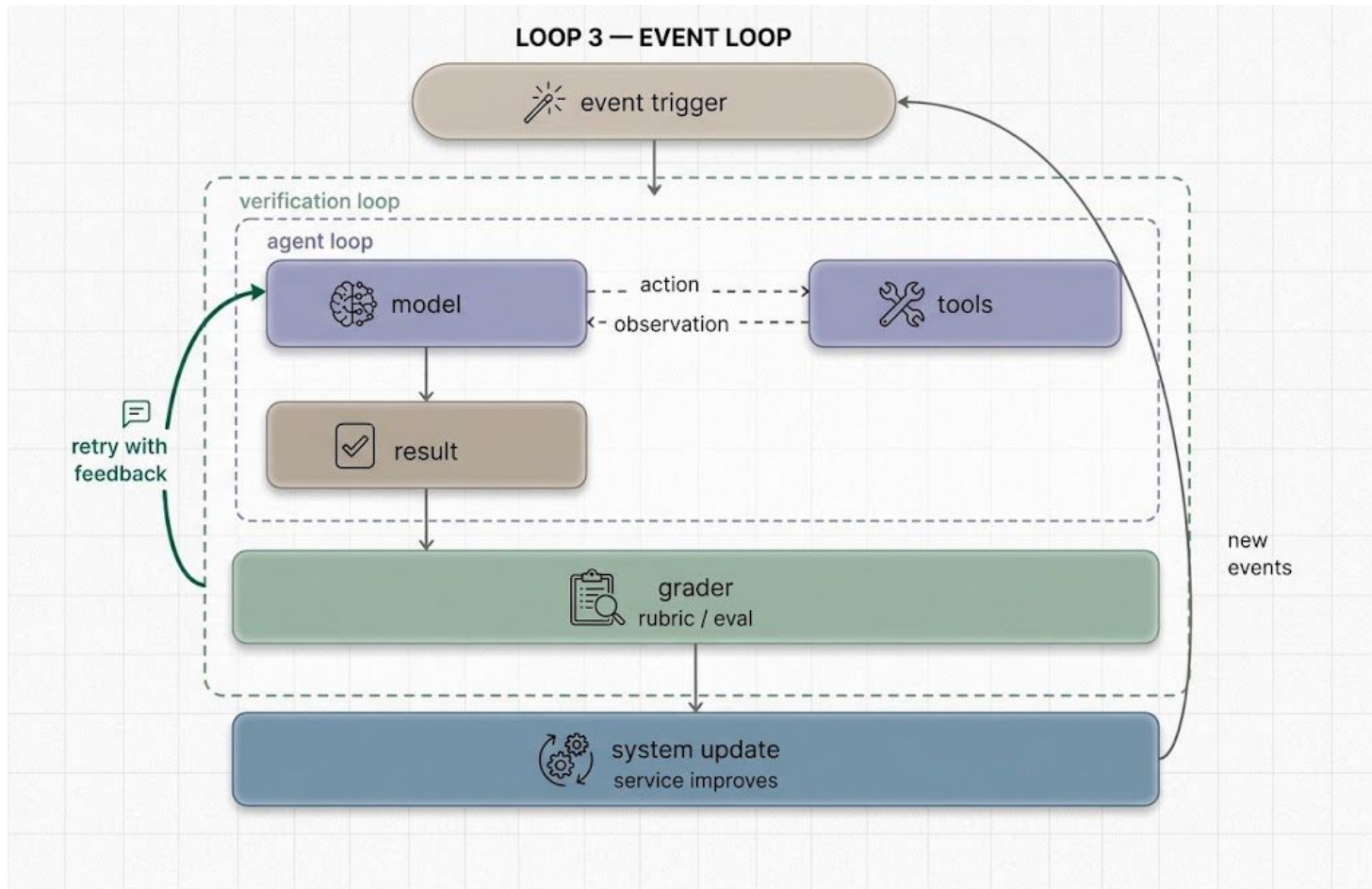
What is a (agentic) loop?

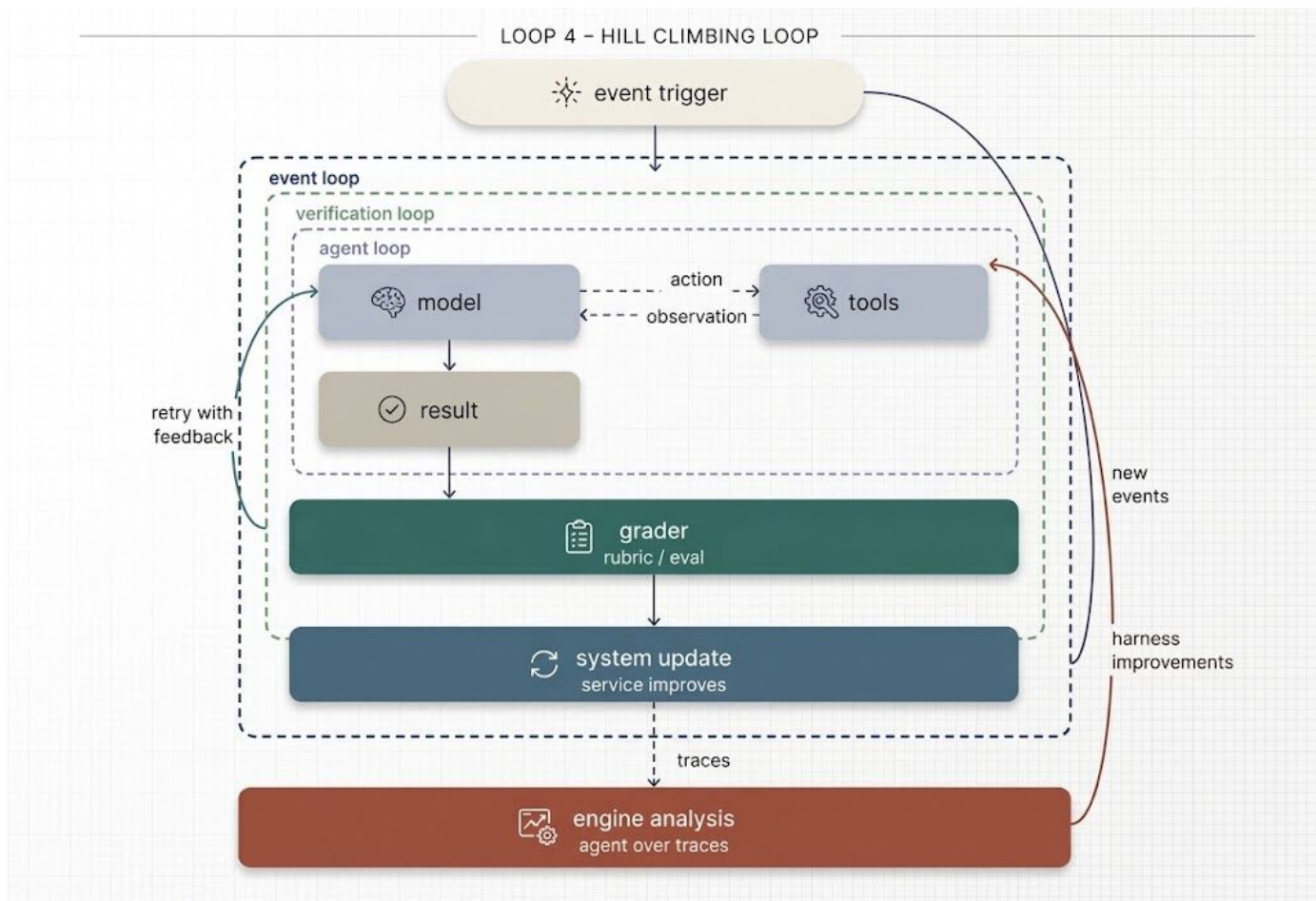
LOOP 1 — AGENT LOOP



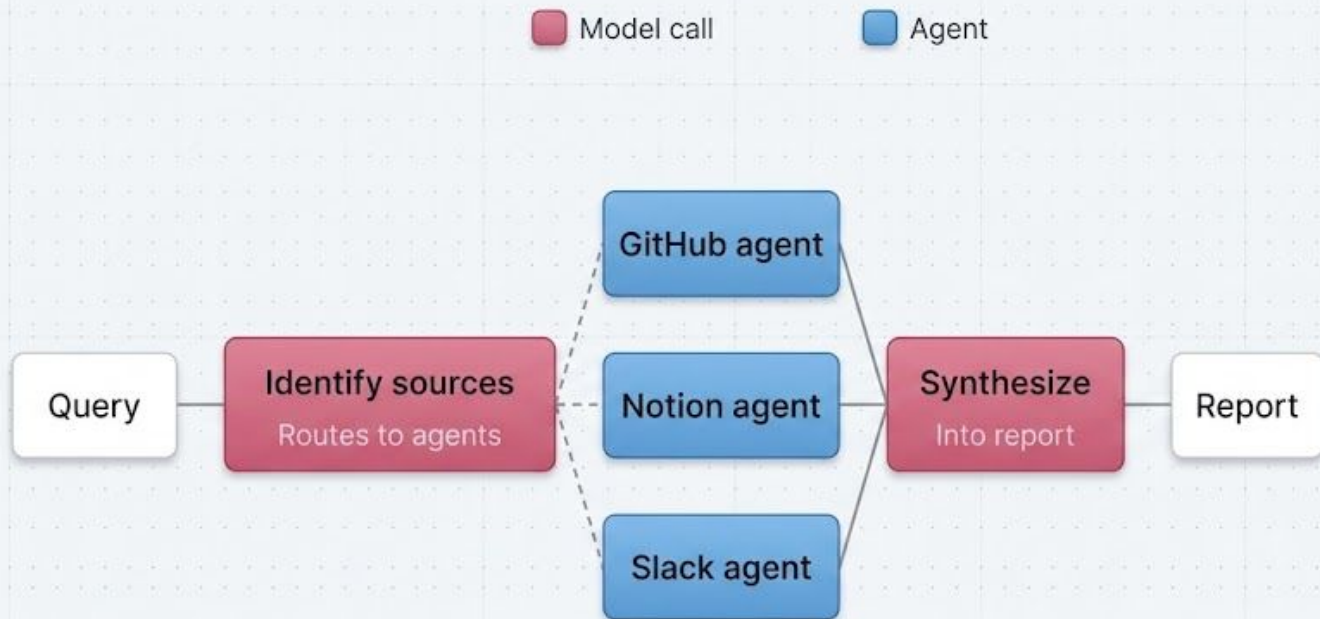
LOOP 2 – VERIFICATION LOOP







What is a (agentic) graph?



We've been doing this a
long time...

The only new part of this.. is the LLM

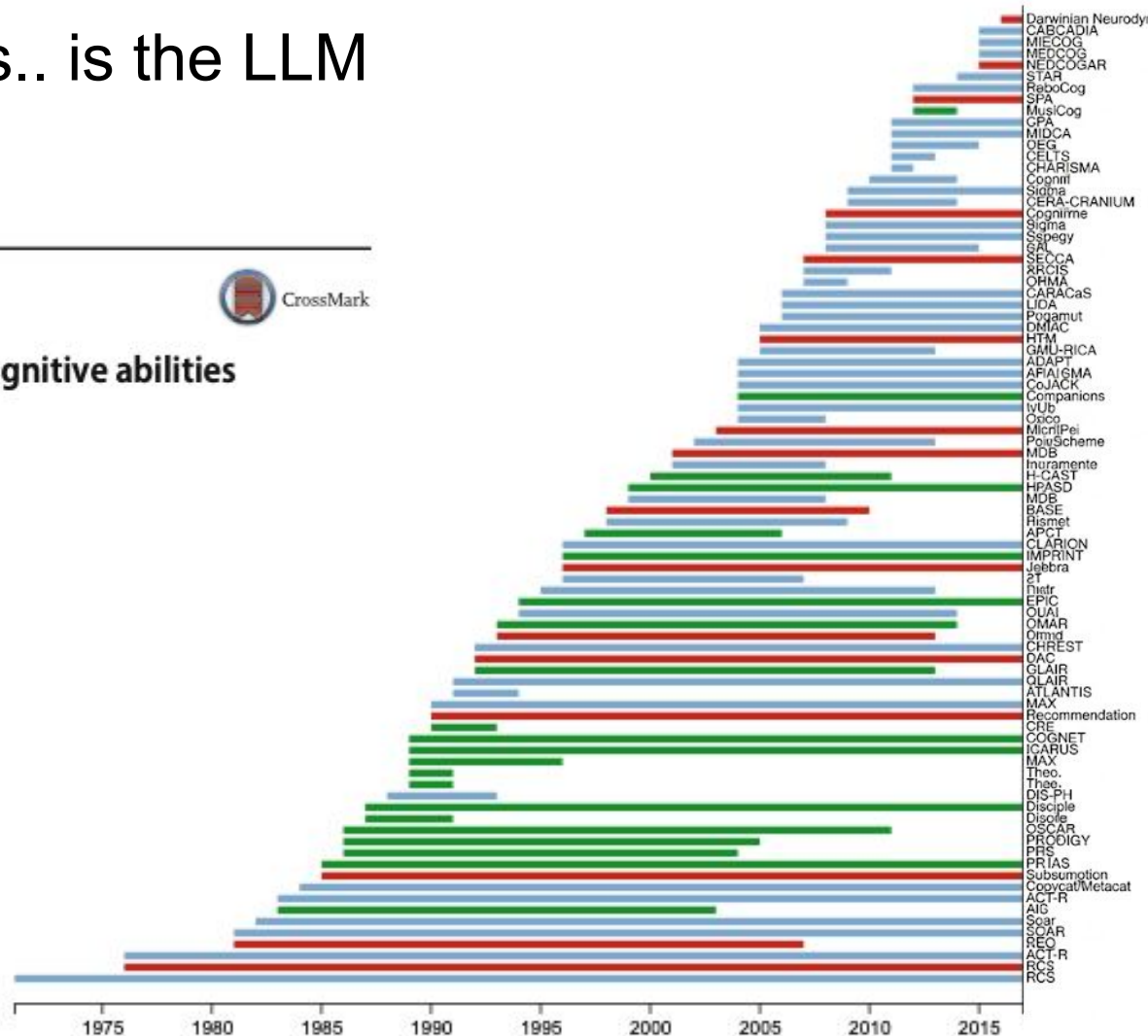
Artificial Intelligence Review (2020) 53:17–94
<https://doi.org/10.1007/s10462-018-9646-y>



40 years of cognitive architectures: core cognitive abilities and practical applications

Iuliia Kotseruba¹ · John K. Tsotsos¹

Published online: 28 July 2018
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The only new part of this.. is the LLM

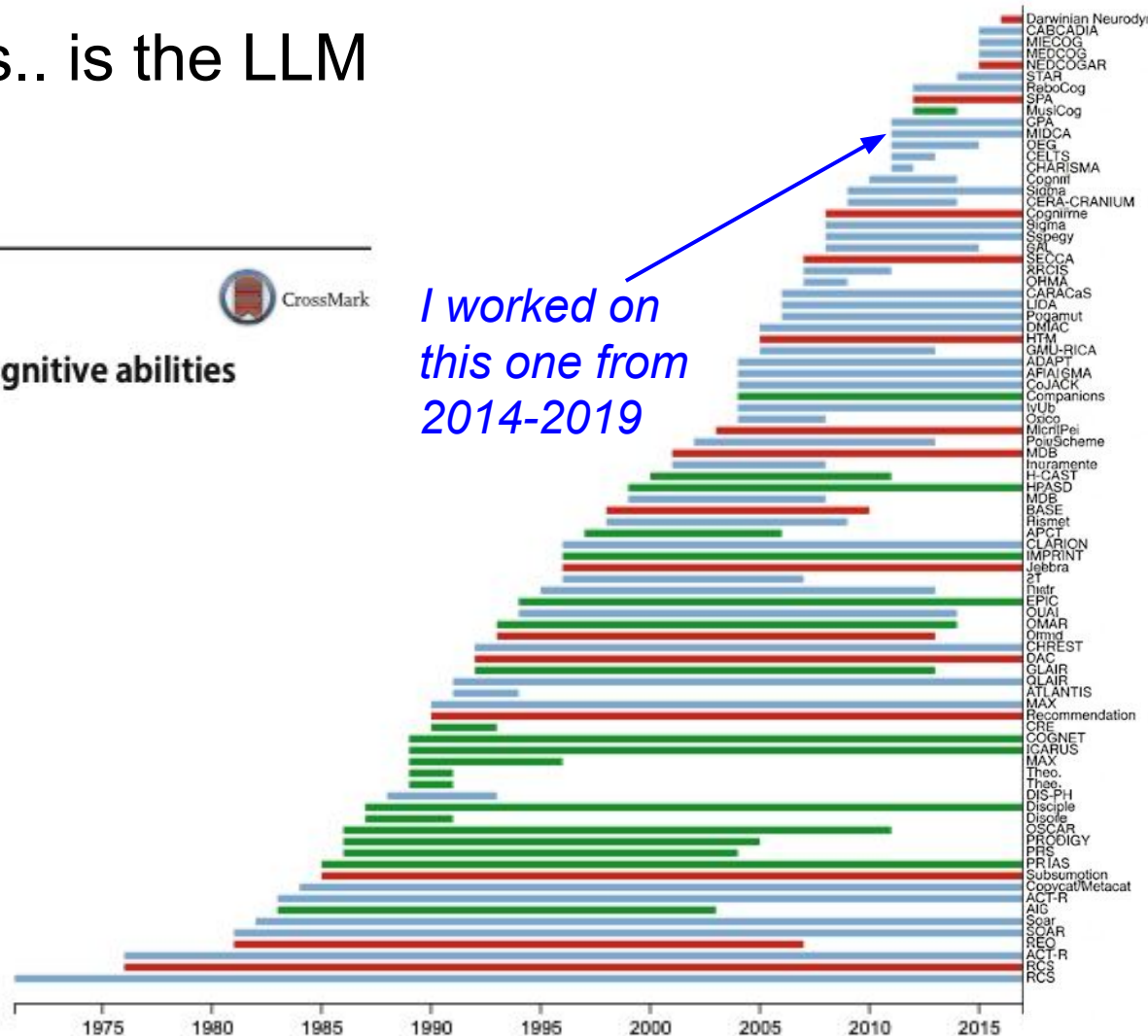
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MIDCA:

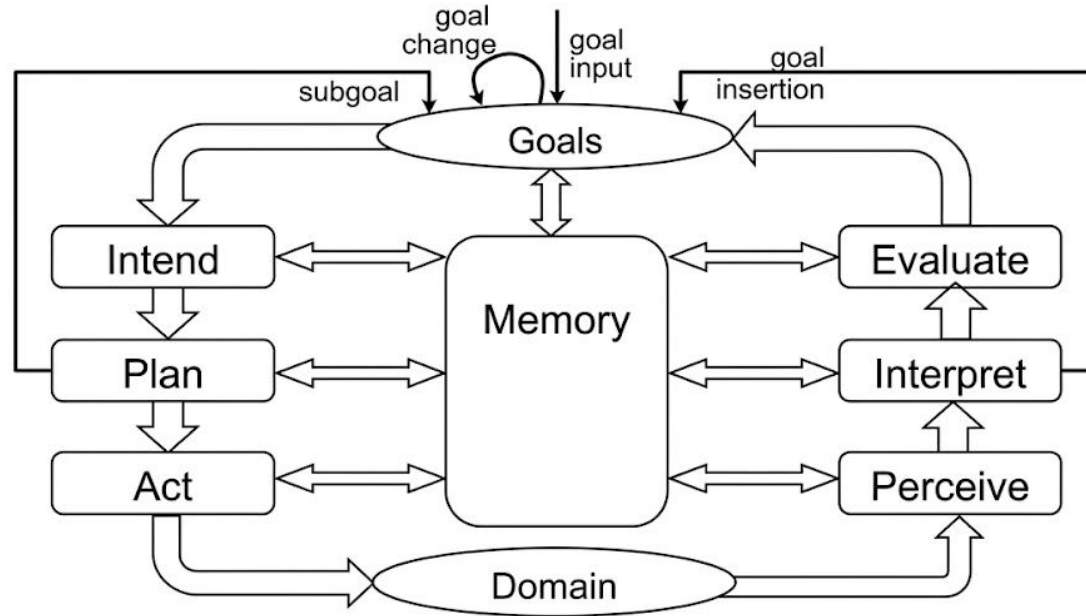
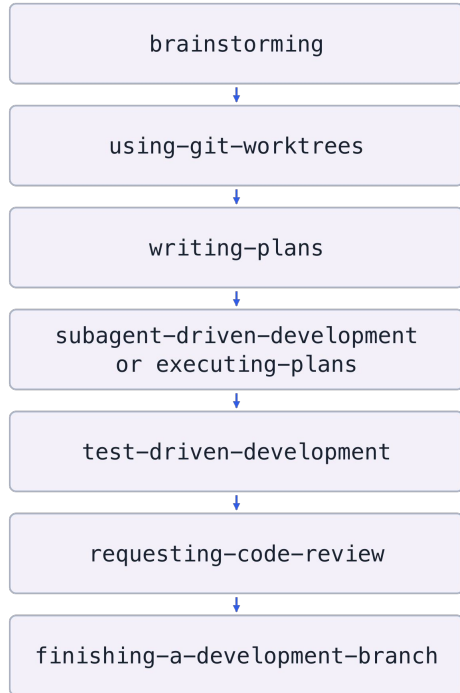
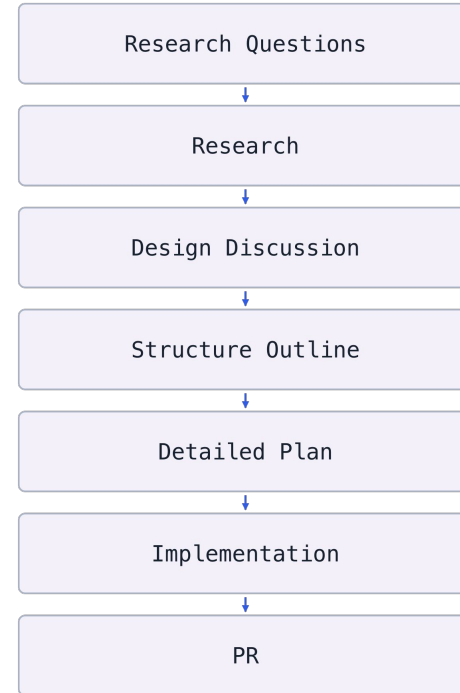


Figure 1. Schematic action-perception cycle for both object and meta-levels

Workflows (i.e. Superpowers, RPI)



github.com/obra/superpowers (283k stars)



github.com/humanlayer/riptide-rpi

Are workflows graphs or loops? Or both?

Workflows usually don't loop, but might branch or have steps that are re-run multiple times.

For example, superpowers workflows are added as skills to harness

If the harness or you don't like the intermediate results, you can re-run or visit a previous step

Unlike many loops/graphs, workflows imply the human being hands on

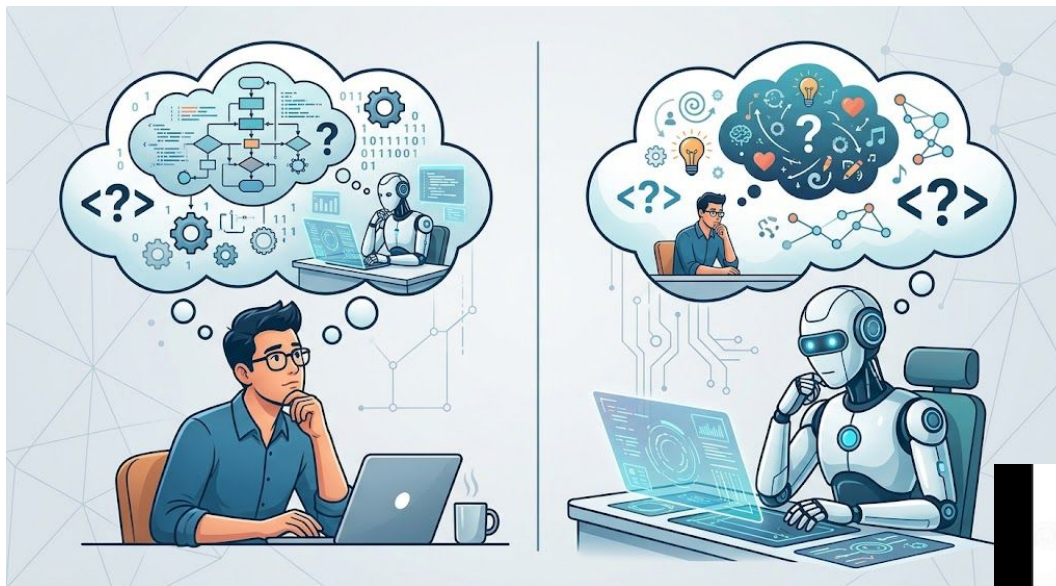
Benefits of Workflows, Loops and Graphs

You can control and test each step independently

You know what process will be followed to do the work

You can develop a mental model of the process

We need good mental models to work together



arXiv

Computer Science > Artificial Intelligence

[Submitted on 28 Jan 2017 (v1), last revised 30 May 2017 (this version, v5)]

Plan Explanations as Model Reconciliation: Moving Beyond Explanation as

Tathagata Chakraborti, Sarath Sreedharan, Yu Zhang, Subbarao Kambhampati



Subbarao Kambhampati

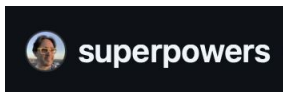
@rao2z

Outline

1. Loops, Graphs, and Workflows
2. **General Approaches to Agentic Coding Work**
3. Software as a Discovery Process
4. Early look at Playbooks

Major Approaches Today

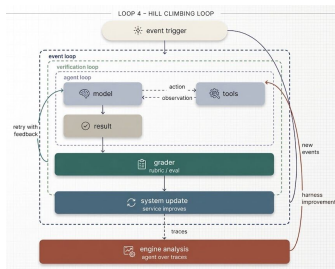
Skill-guided Coding



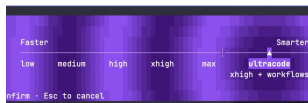
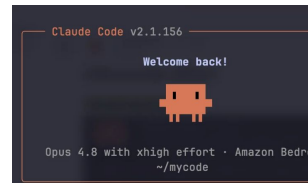
Workflows



Programmed Loops and Graphs



Dynamic Loops and Graphs



/goal

/goal

- **You define the outcome and completion criteria**
- **The agent chooses the steps as it goes**
- **Work, check results, and adjust the approach**
- **Automatically continues across turns**
- **Designed for largely hands-off execution**
- **Runs until completion or a stopping condition**

Dynamic Workflows

- **Claude designs the process** for your task
- **A generated script coordinates agents**
- **Parallel work, branches, and verification loops**
- **Intermediate results stay outside the main conversation**
- **Long runs with limited human involvement**
- **Inspect, edit, and save workflows for reuse**

Task Dimensions

Dimensions

Dimension	0 means...	100 means...
Human Worker	Difficult to have a human be part of the process	Work can be handed to humans just as easily as it can be given to AI

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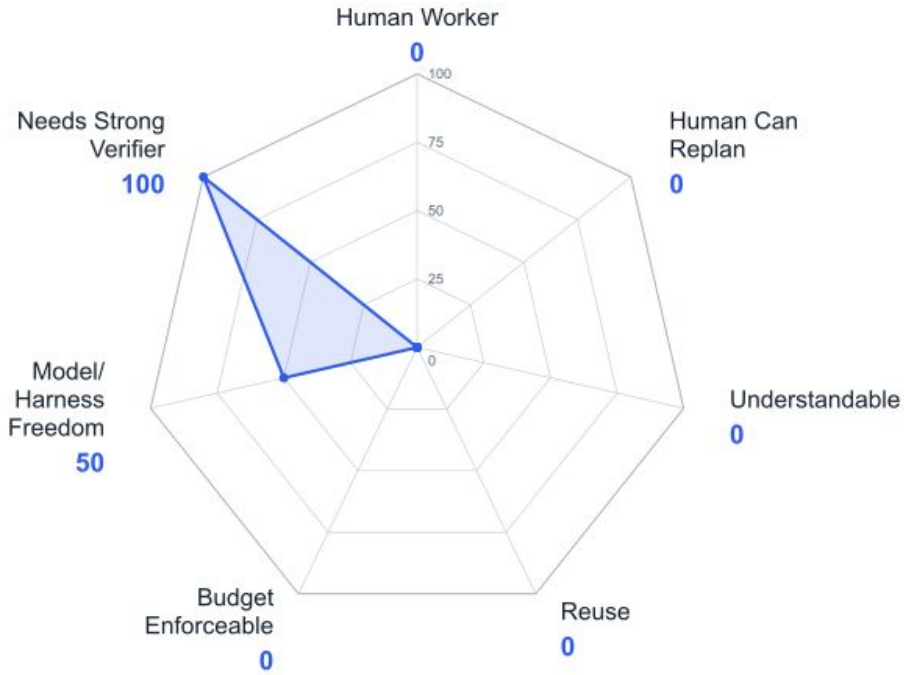
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Budget Enforceable	No ability to influence how much it will cost	Easy to enforce token, time, money budgets

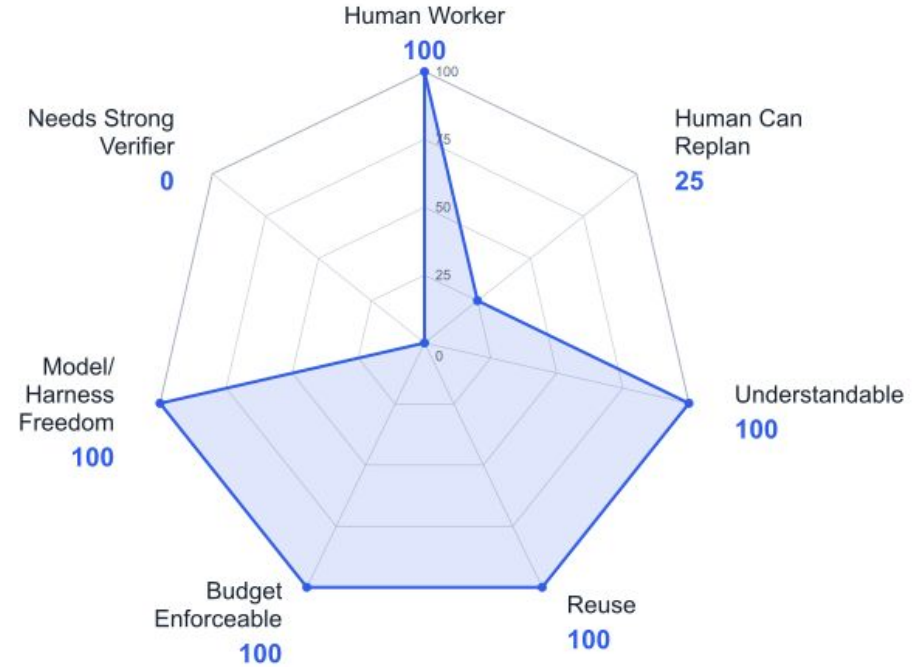
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Understandable	Extremely difficult for human to form mental model	Easy to form a mental model

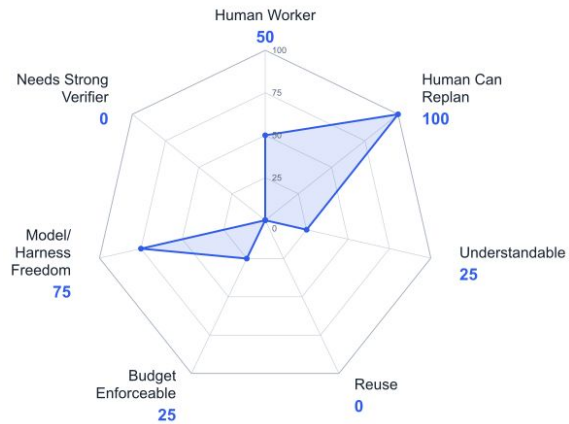
/goal



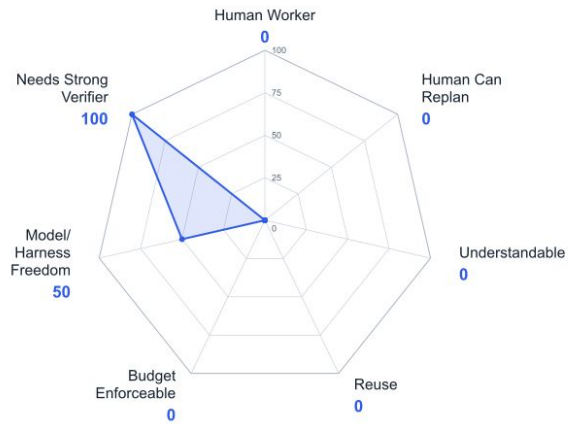
Programmed Loops and Graphs



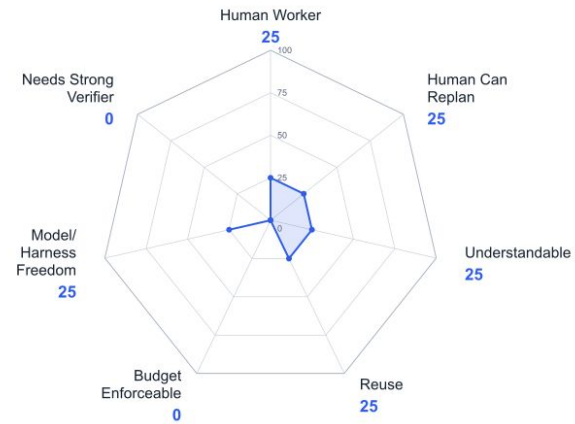
Skill-guided Coding



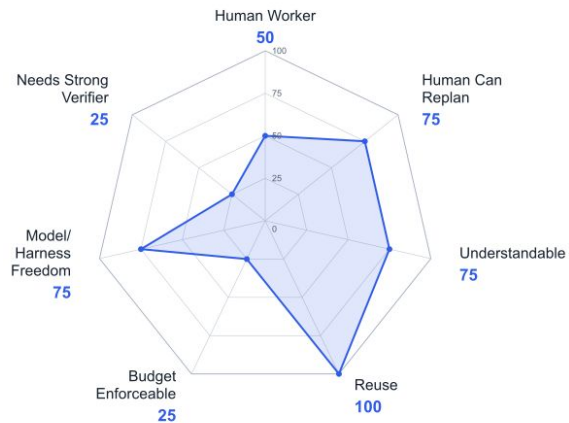
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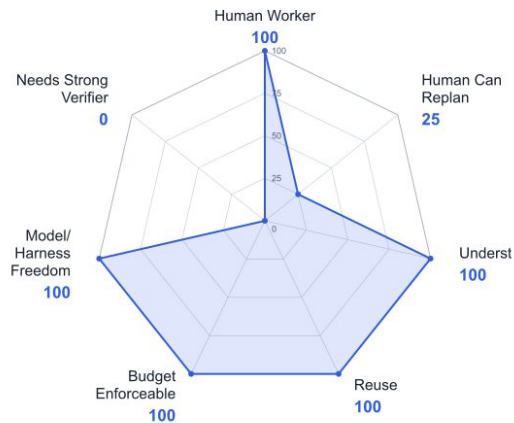
Dynamic Workflows



Workflows



Programmed Loops and Graphs



New model release, maybe ASI is here?

I tried this Day 2 of Astra being released

- Major new feature: “Designing an engine to execute playbooks”
- I answered ~200 questions over two days, lots of back and forth
- Ran Astra **/goal** and used > 700M Tokens in 3 hrs
- It wrote 32000 LOC and removed 48000 LOC

Environment	+
Changes	+31,238 -48,821

How do you think it went?

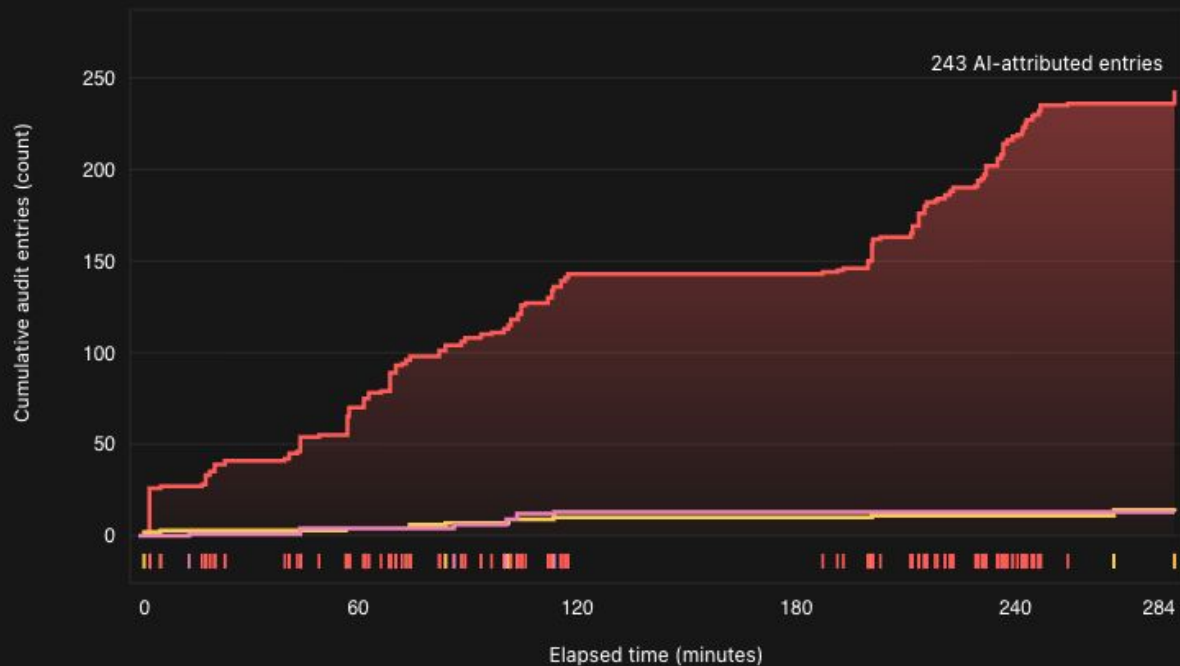
Choices inside the one-shot run

Audit timeline

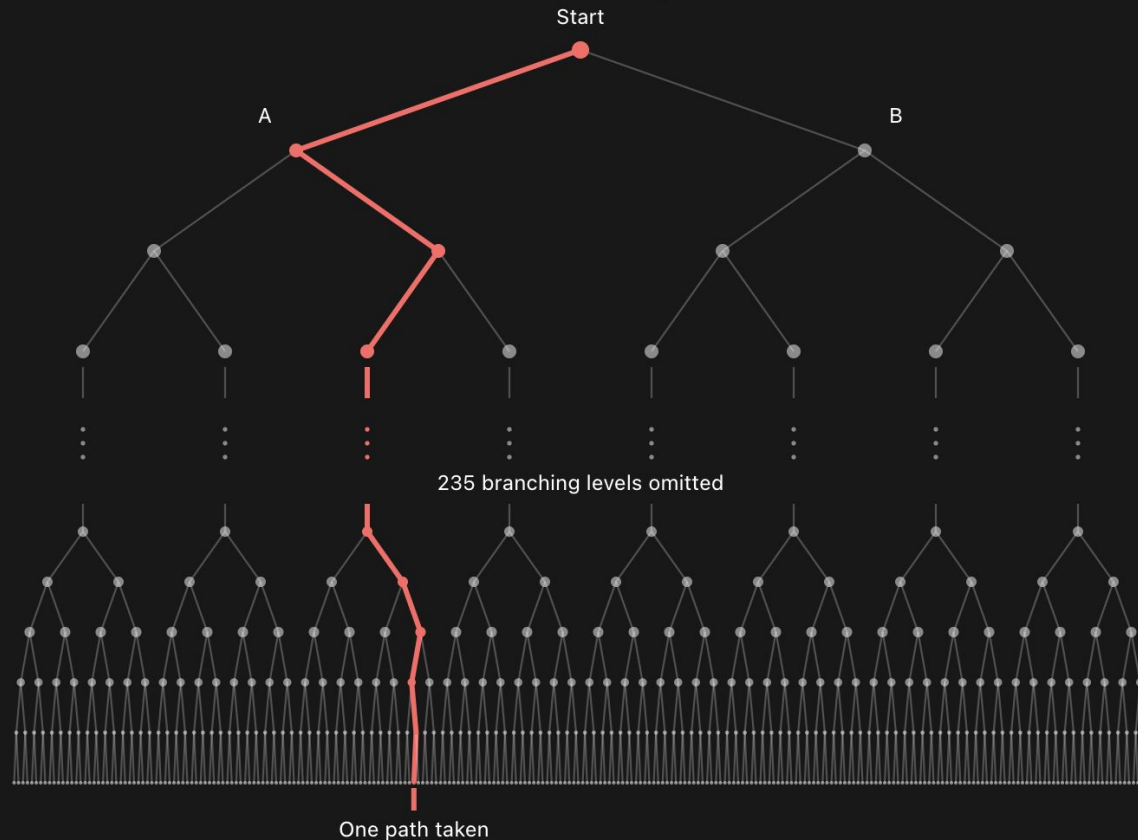
Possibility model

— 243 AI-attributed — 15 Inherited — 13 Unclear

271 extracted decision entries · 311 assistant messages · 4h 44m



243 decisions. One path taken.



$\approx 1.41 \times 10^{73}$ possible paths

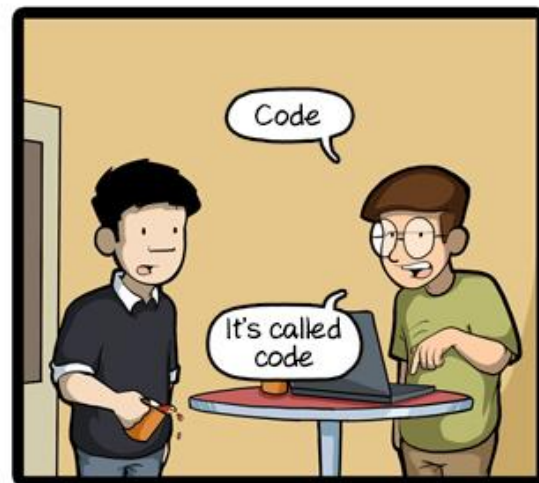
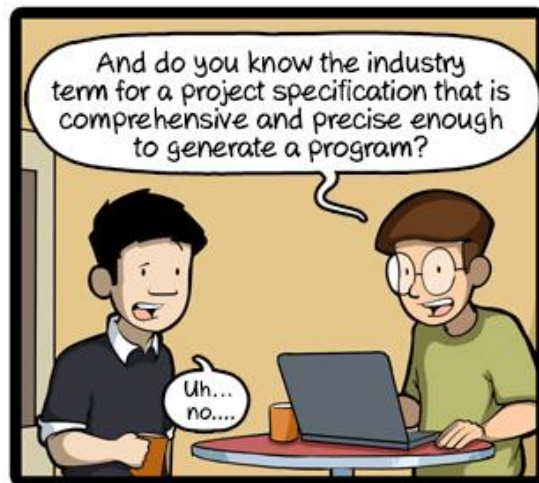
Two choices at every fork · 256 illustrative leaves shown

A few really bad decisions early on, snowballed from there

1. Every task gets it's own database
2. An 16GB sparse APFS disk image for every Execution, with unmount/remount during completion.

This lead to bad downstream effects:

- Inside SQLite, it mostly stores JSON and reconstructs the database's answers in Rust.
- Cross-Task relationships lose ordinary foreign-key enforcement
- Listing, backup, and recovery all have to coordinate separate stores.



Software Development is a Discovery Process

You start with an idea

There are many restrictions from hardware, software, programming languages, algorithms, etc that you don't know (and the AI doesn't either) until you have code

You're really working to find a compromise between what you want and what is possible to specify in code.

Every feature is another compromise, so it gets really complex! When you have an idea, you don't yet know what the tradeoff space is and whether you are willing to accept the tradeoffs involved in what you want!

Frontier models are not holding you back

It's your process for coding!

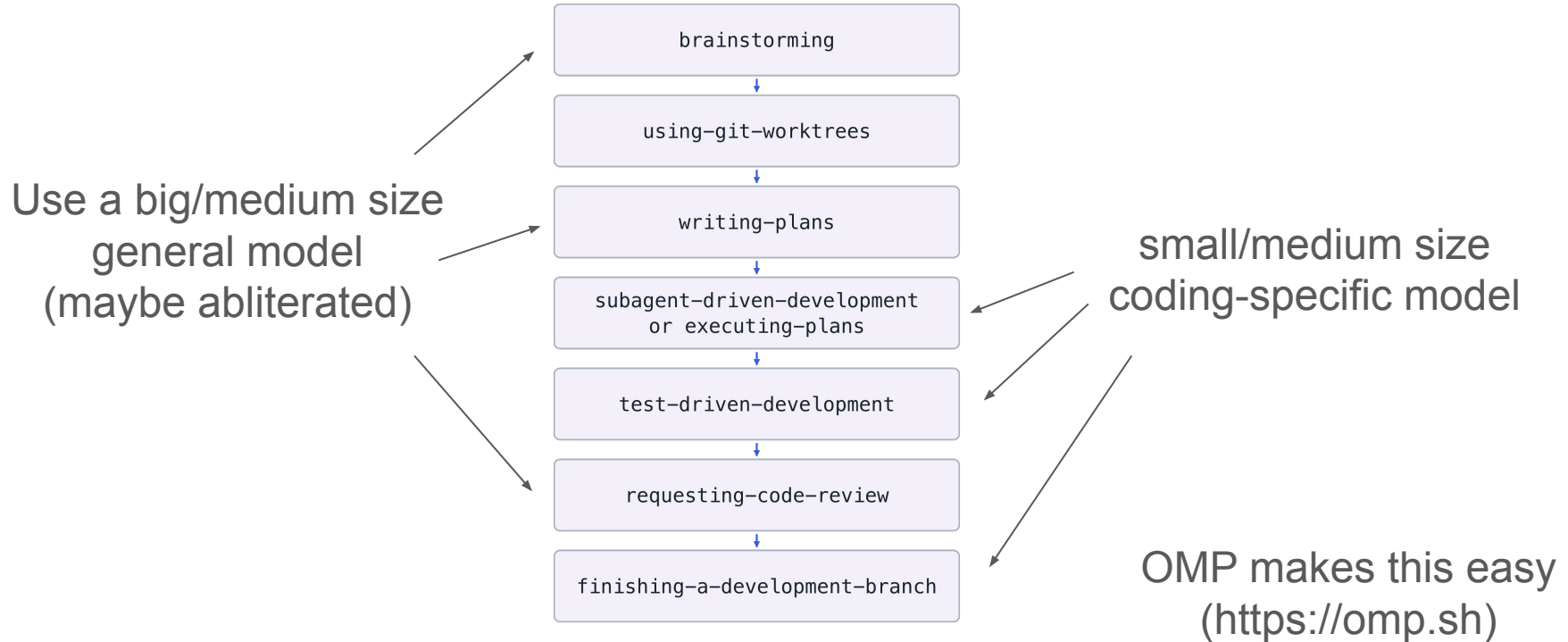
And this is great news for open source AI and local AI

Use more workflows, loops, graphs, and playbooks!

You don't need to pay a Frontier Lab to get Frontier Results

**Great News for
Open Models!**

How to use local models with workflows



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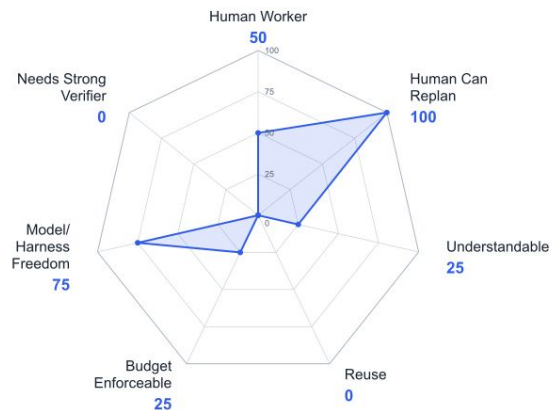
Playbooks

What are Playbooks?

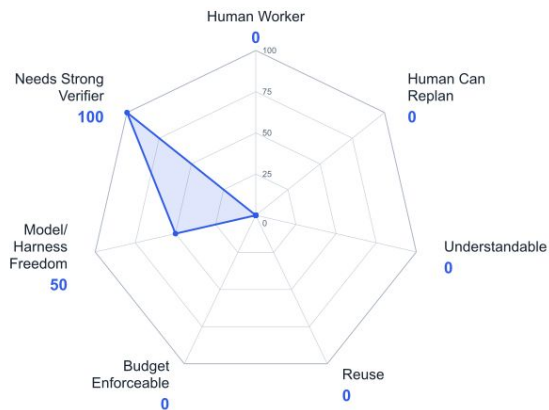
A playbook is if I want to start with a team of agents who already know what to do, what the play is.

And I want them to be taught by an expert

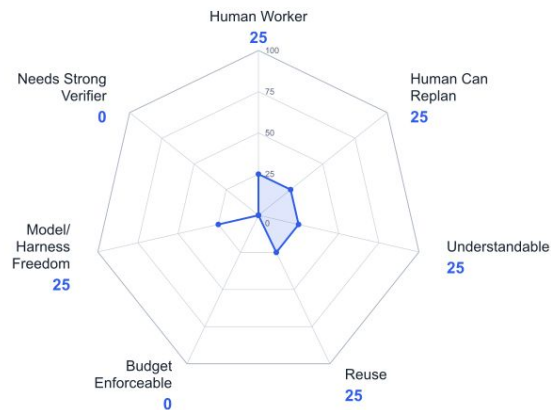
Plain Harness



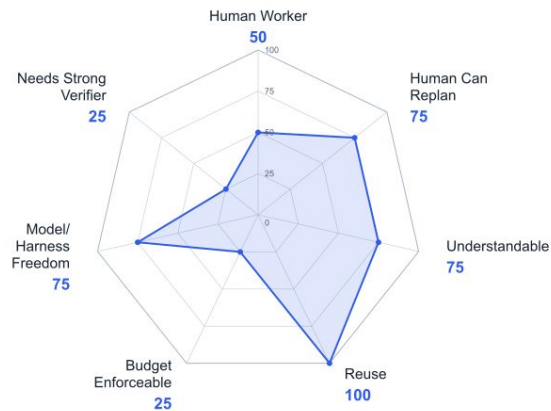
/goal



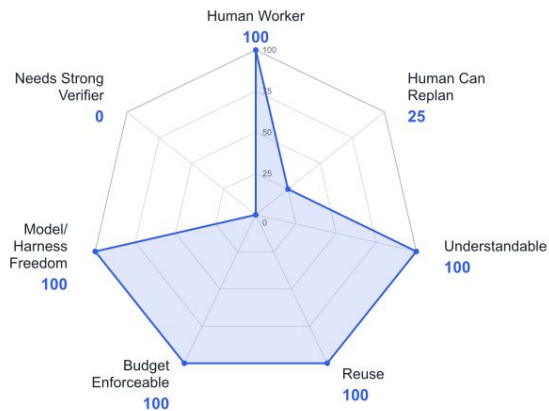
Dynamic Workflows



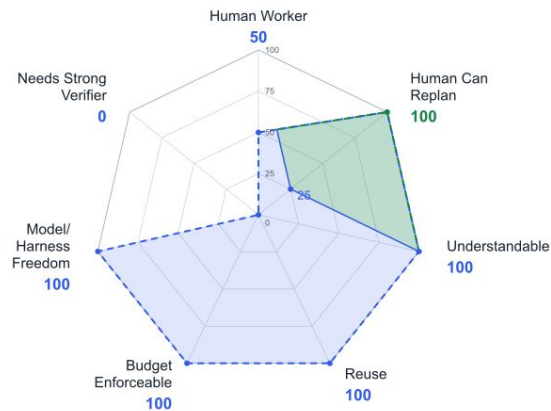
Workflows



Programmed Loops and Graphs

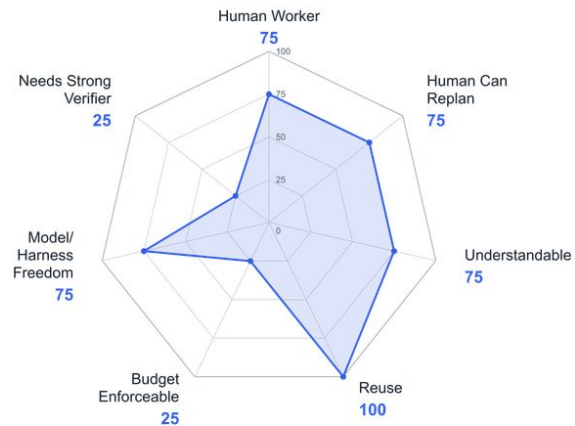


Playbooks + Alinery

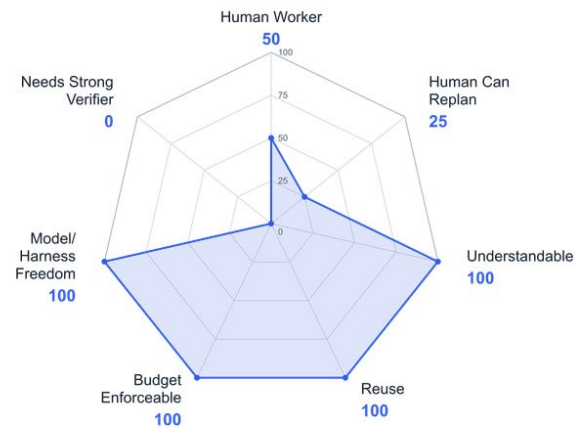


■ Additional human replanning: 25 → 100

Workflows



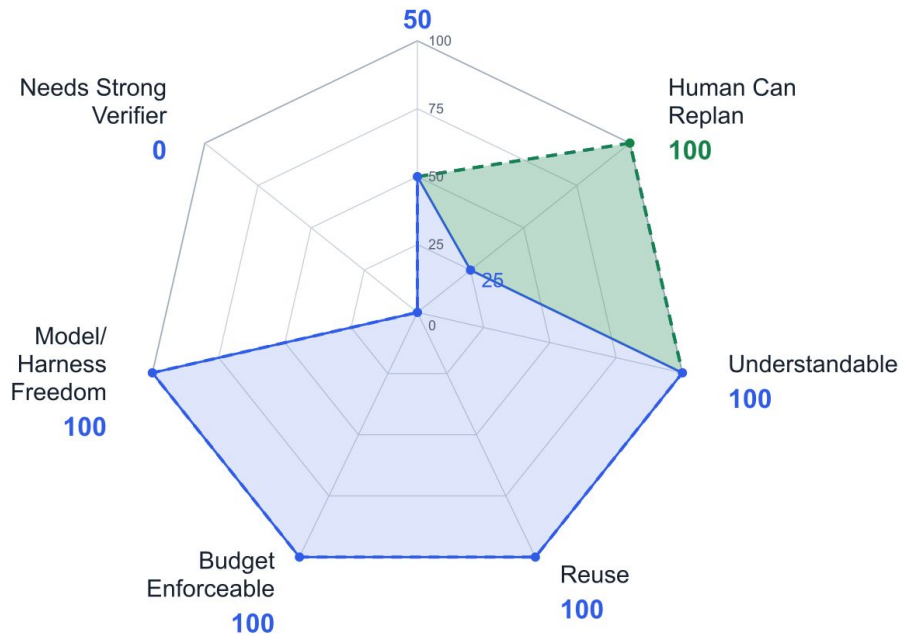
Programmed Loops and Graphs



Playbooks

DESIGN TARGET

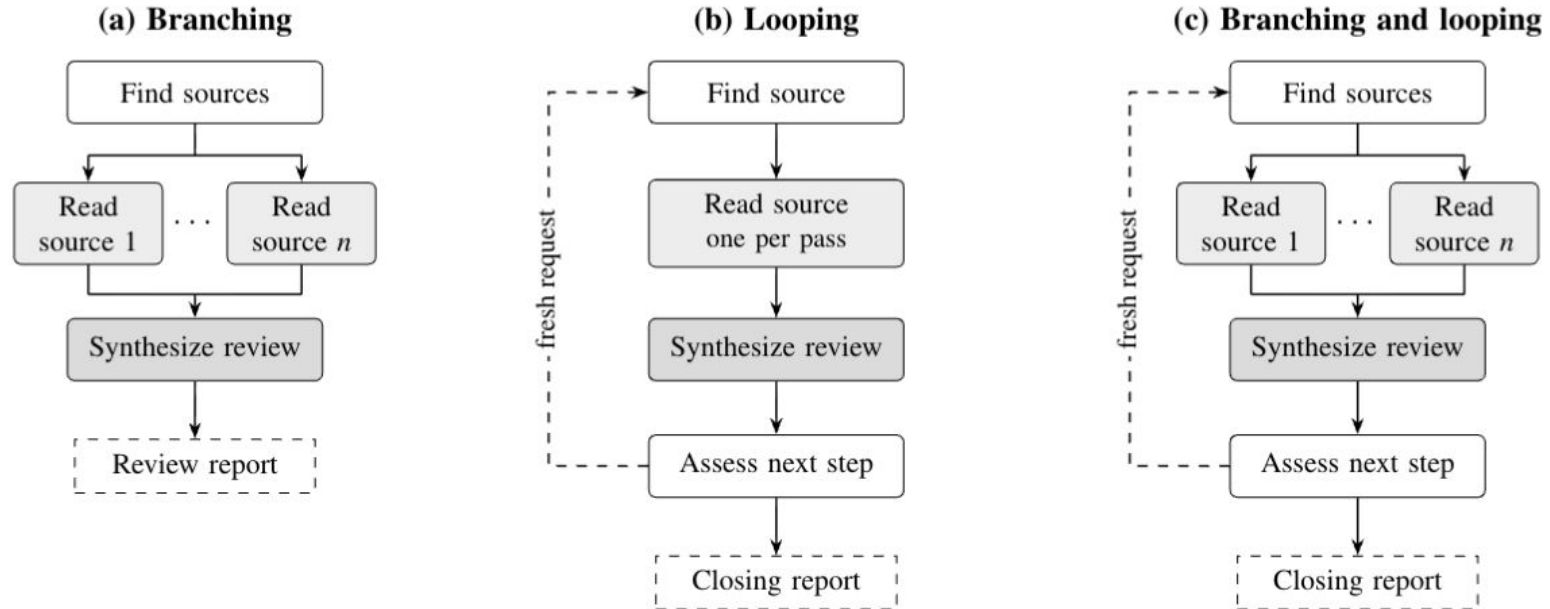
Human Worker



Shared with Programmed Loops and Graphs

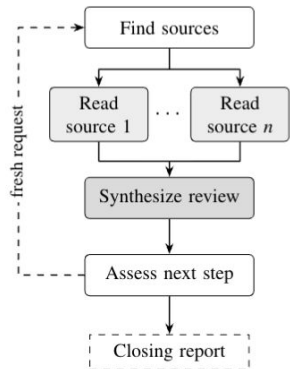
Added Human Can Replan: 25 → 100

The Power of Rollouts

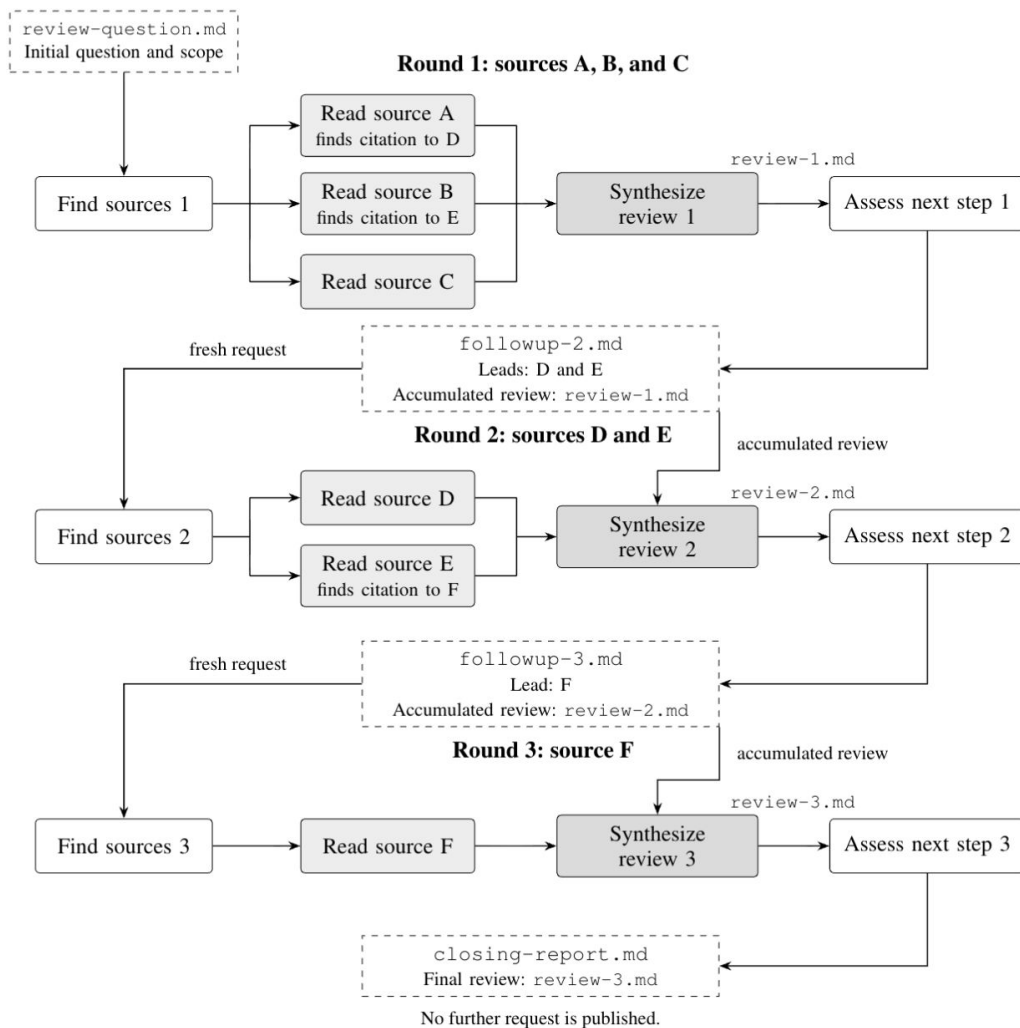


The Power of Rollouts

(c) Branching and looping



executes as...



Create Playbooks

Repository

Alinery Review

Bind an Alinery change, run safe checks, inspect code, vision, design, website claims, and performance, then prepare a paste-ready review.

► Source details

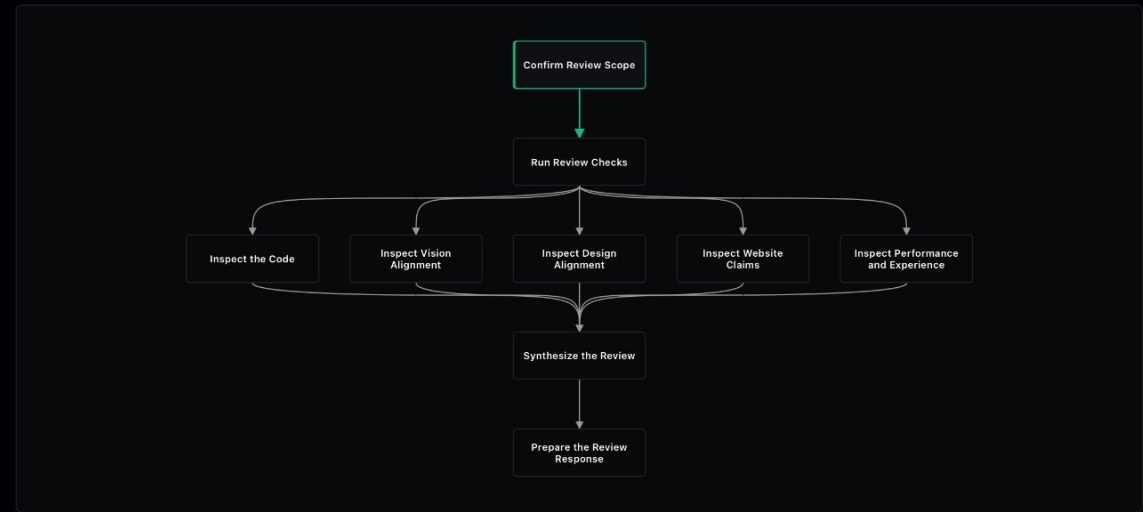
Flow Artifact dependencies

-

57%

+

Fit graphReset zoomPan



Confirm Review Scope

Full artifact dependencies, including connections omitted from Flow.

Key	bind
Title	Confirm Review Scope
Short label	bind
Model	Inherits task / product default
Harness	Inherits playbook default: omp
Coding step	No
Automatic completion by default	Yes

Inputs

ticket.md single

Outputs

review-context.md

► Connections (8)

Saved definition

Delete Playbook

Run Playbooks

Task

review pr on notifications / status

Branch review-pr-on-notifications-status

Worktree /Users/dustin/Projects/Public/alinery/.alinery/worktrees/review-pr-on-notifications-status

PR URL not available

Related Tag a related task...

Duplicate Task · #D Archive task

Sessions

List

0 queued Show archived Start sub-task New session

57% + Reset view Pan

2-review-checks-1.md Copy Preview Raw

Review checks — PR #52

Result

Runtime/build verification blocked by a pre-existing target mismatch. The existing worktree is clean but does not contain the frozen PR head. No product checks were run against the unrelated checkout. This report completes the checks execution's evidence collection; it is not a passing verification result or a recommendation to merge.

Inputs and frozen target

Read only the assigned ticket and review-context input occurrences, plus applicable supplied repository instructions:

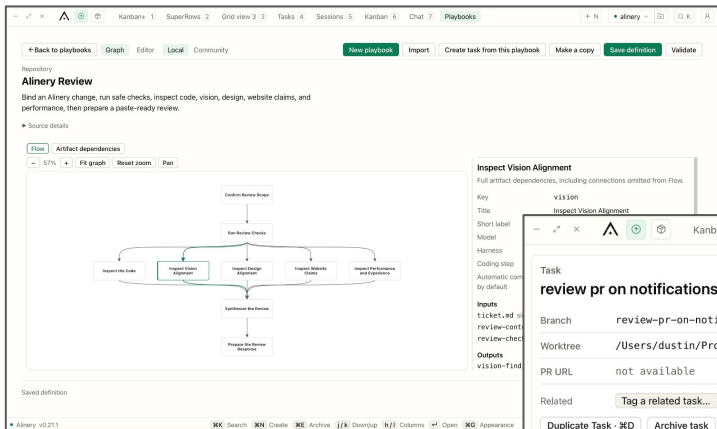
- Ticket: 00-ticket.md, occurrence d7dae57d-18e3-40bf-b497-3733f0be1297 .
- Review context: 1-review-context-1.md, occurrence 6d9cf88f-88c5-472a-b3b8-d9ce7007261a .
- Request: review https://github.com/DeleganceAI/alinery/p/notifications/status changes.

Alinery v0.21.1

#K Search #E Archive #D Duplicate esc Back

Name ↕	Playbook ↕	Active session ↕	Status ↕	Sessions ↕	Created ↕	Updated ↕
review pr on notifications / status review-pr-on-notifications-status	Alinery Review	—	🟢	11	1w	now
review chat pr review-chat-pr	Alinery Review	Prepare the Review Response	👉	9	1d	now
if human starts talking, don't auto advance by default if-human-starts-talking-don-t-auto-advance-by-default	SuperDevelop	—	🟢	8	1w	now
on new session page, need option to choose available next steps on-new-session-page-need-option-to-choose-available-next-steps	SuperDevelop	—	🟢	8	5d	now
updated column stale updated-column-stale	SuperDevelop	—	🟢	8	5d	now
DEL-724 — Exited chat remains writable and shows Working codex/del-724-exited-chat-remains-writable	SuperDevelop	—	🟢	5	1w	now
DEL-723 — Older history duplicates live messages D+1 codex/del-723-older-history-duplicates-1	SuperDevelop	—	🟢	5	5d	now
create solution design loop playbook create-solution-design-loop-playbook-1	Build a New Playbook	—	🟢	5	5d	now
ask tool with omp ask-tool-with-omp	Bug Hunting	—	🟢	4	5d	now
Investigate Playbook notification and status-update wiring gaps investigate-playbook-notification-and-status-update-wiring-gaps	Bug Hunting	—	🟢	3	1w	now
picking back up on multiple sub tasks picking-back-up-on-multiple-sub-tasks	One-shot	—	🟢	3	1w	now
create a whitepaper paper writing playbook create-a-whitepaper-paper-writing-playbook	Build a New Playbook	—	🟡	3	2d	now
show pr status on task list page show-pr-status-on-task-list-page	One-shot	—	🟢	2	1w	now
new conference review playbook new-conference-review-playbook	Build a New Playbook	—	🟡	2	1d	now
change one-shot to be 1 step playbook	One-shot	—	🟢	1	1d	now

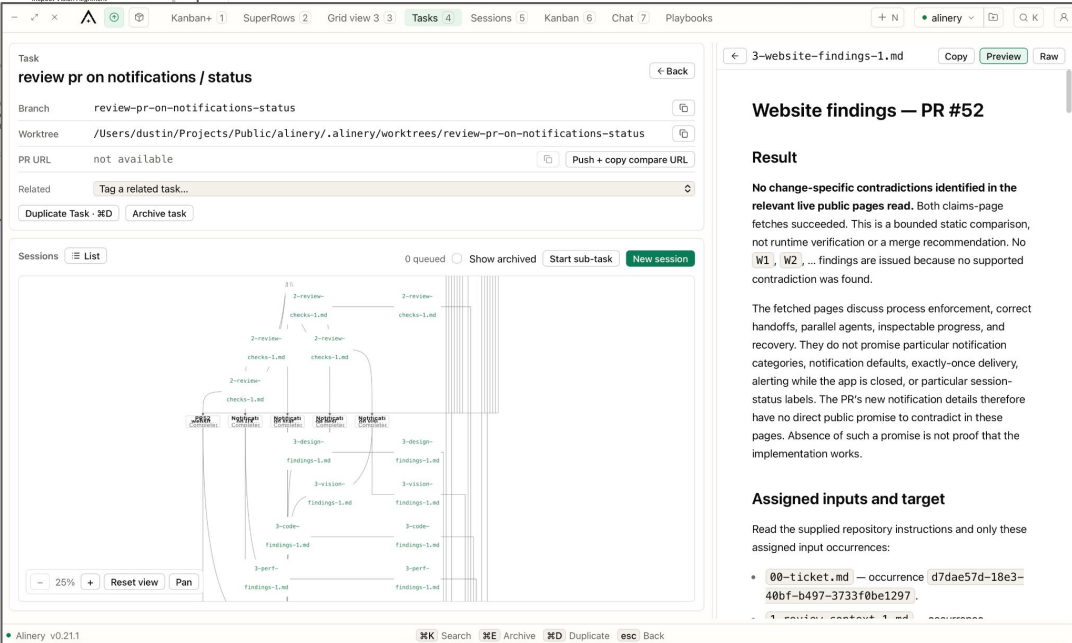
Light Mode



Tasks

New task ☐ Show archived

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review pr on notifications / status review-pr-on-notifications-status	Alinery Review	—		11	1w	now
review chat pr review-chat-pr	Alinery Review	Prepare the Review Response		9	1d	now
if human starts talking, don't auto advance by default if-human-starts-talking-don-t-auto-advance-by-default	SuperDevelop	—		8	1w	now
on new session page, need option to choose available next steps on-new-session-page-need-option-to-choose-available-next-steps	SuperDevelop	—		8	5d	now
updated column stale updated-column-stale	SuperDevelop	—		8	5d	now
DEL-724 — Exited chat remains writable and shows Working codex/del-724-exited-chat-remains-writable	SuperDevelop	—		5	1w	now
DEL-723 — Older history duplicates live messages D+1 codex/del-723-older-history-duplicates-1	SuperDevelop	—		5	5d	now
create solution design loop playbook create-solution-design-loop-playbook-1	Build a New Playbook	—		5	5d	now



Live Demo App to Check Your Process + Skill



I want to...

build an end to end encrypted video calling app

Find the optimal AI process

GPT-6-Luna

Using ChatGPT plan Manage usage

GPT-6-Luna · ~1,700 input tokens + reply/reasoning · call 1 of 2

[► About this estimate](#)

Sent to OpenAI using your ChatGPT plan when you submit. Not saved on this server; provider retention may apply.

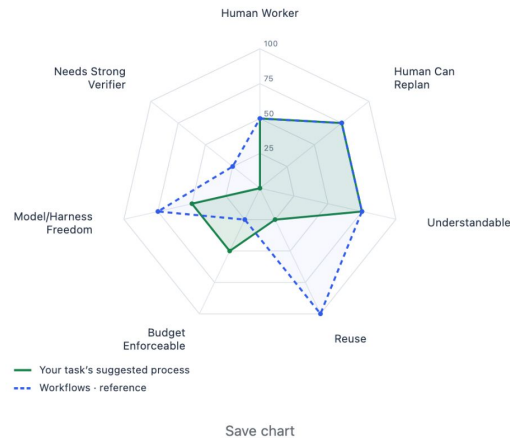
3 / 3 · YOUR STARTING POINT

Workflows

Start with a lightweight, reusable sequence such as requirements and threat modeling → architecture review → implementation → tests and independent security review. Phase boundaries make it easier to inspect assumptions and adapt the design without requiring a maintained graph or a strong automatic success signal. Keep an experienced security reviewer involved before consequential use.

How to get started

1. Clarify intended users, platforms, and prototype scope; keep real private messages and credentials out of prompts, logs, and test fixtures.
2. Write down a threat model and security assumptions, then arrange experienced security review of the proposed design before relying on it.
3. Use short workflow phases with human checkpoints for requirements, design, implementation, and review; revise the plan when assumptions change.
4. Use realistic functional tests and manual review, while recording explicitly what those checks do not establish about security.
5. Do not use the prototype for sensitive communications unless qualified review supports that use and remaining risks are understood.



[Save chart](#)

Alinery is our open source Playbook IDE

- Develop with Playbooks
- Organize many agents per task
- Explore rabbit holes via sub-tasks
- Import from Github/Linear
- ... and much more soon!

Discord: <https://discord.gg/tgAUKEv5U>

Website: <https://alinery.ai>

Github: [DeleganceAI/alinery](https://github.com/DeleganceAI/alinery)

Drop me a line at dustin@delegance.ai

